

# Aircraft Sequencing Problems via a Rolling Horizon Algorithm

F. Furini, C. Persiani, P. Toth

Dipartimento di Elettronica, Informatica e Sistemistica, University of Bologna,  
Viale Risorgimento, 2 - 40136 - Bologna (Italy)  
{fabio.furini, alfredo.persiani, paolo.toth}@unibo.it

## Abstract

*Aircraft sequencing on the runway is a challenging optimization problem that aims to reduce the delays and the air traffic controllers workload in a scenario characterized by a continuous growth of the air transportation demand. In this paper we consider the problem of sequencing both arrivals and departures on a single runway airport. We formalize the problem using a Mixed Integer Programming Model and we propose a rolling horizon solution approach. Computational results on real-world air traffic instances from the Milano Linate Airport are reported. The results show that the proposed approach is able to significantly improve on the First Come First Served (FCFS) sequence.*

**Keywords:** Aircraft Runway Scheduling, MILP, Rolling Horizon

## 1 Introduction

At busy airports the runways are used by thousands of aircraft every day. During peak hours the pre-departure sequence is often established using the First Come First Served (FCFS) principle and the Air Traffic Controllers assign a delay on the start-up operations. At the same time many arrivals enter the airspace via the entry Fix (entry point of the terminal area of the airport) and they are sequenced on the ILS (Instrumental Landing System) path by the Air Traffic Controllers. The arrival sequence is established considering several factors, including aircraft type and performance (Ground Speed and Rate of descent), their position and distance from the ILS path and the workload required for the Controllers to define a sequence. The arrivals and the departures sequences are integrated tactically by the Tower Controllers, for instance a departure is inserted between two arrivals once the Controller is sure that the appropriate separation will be provided. Few airports in Europe use Arrivals Management (AMAN) systems and/or Departure Management (DMAN) systems. These tools provide important information to the Controllers and suggestions on the aircraft sequencing. Currently AMAN and DMAN systems operate separately, and many research projects, such as SESAR (Single European Sky Air traffic management Research), are dealing with the integration of these systems. In this context, sequencing arrivals and departures on a runway is a challenging problem that will become crucial for future air traffic management.

The literature on runway scheduling problems is quite varied. Even though some authors mention how to extend their approaches to a mixture of arrivals and departures, the landing and take-off scheduling phases are generally considered as separate problems: the Aircraft Landing Problem (ALP) and the Aircraft Take-Off Problem (ATP). A complete overview on the techniques and tools used for the landing and take-off scheduling problems has been recently presented by Bennel et al. [6]. Basically two versions of the aircraft sequencing problem have been considered in the literature: the static version where it is assumed that the complete information of a given set of aircraft are known, and the dynamic version where the solutions are revised over the time as new aircraft arrive into the system. One of the most

interesting mathematical formulations of the static ALP as a mixed-integer zero-one program has been proposed by Beasley et al. [11]. Valid inequalities able to strengthen this formulation have been proposed and an exact algorithm based on a linear tree search has been designed. Both single and multiple runways are considered in the formulation. The dynamic case has been considered by Beasley et al. [12] as an application of the displacement problem. An earlier mixed integer linear programming formulation has been presented by Abela et al [5]: a Branch-and-Bound and a genetic algorithm have been proposed to solve the problem. The considered objective function has a cost for each aircraft which increases its speed or performs holdings.

The Genetic Algorithms represent the most used metaheuristic approach for the solution of the ALP. One of the first applications has been proposed by Stevens [13]. Ciesielski and Scerri [14] presented a genetic algorithm able to solve the real time case of the landing scheduling problem. The algorithm has been tested through a series of experiments on landing data for the Sydney Airport on the busiest days of the year. Four different genetic algorithms have been also proposed by Cheng et al [4]. Beasley and Pinol [15] considered the multiple runways case and proposed a scatter search approach and a bionomic algorithm. They also presented a mathematical formulation with two types of objective functions: a linear one and a nonlinear one. Atkin al. [8] presented and compared a Tabu Search, a Simulated Annealing and a Descent Algorithm for the ATP considering traffic data from the London Heathrow airport. The same authors proposed further developments on the algorithms for the ATP in [7] , [9] and [10].

A model applicable to a mix of take-offs and landings on the same or on different runways has been proposed by Ernst et al. [1]. A branch and bound and a local search heuristic have been designed for the problem. The objective function considers penalty costs for landing before and after the target time. The combination of arrival and departure sequences has been also considered by Persiani and Bagassi [3]. A Particle Swarm Algorithm has been designed to integrate the AMAN and DMAN systems on a single runway busy airport.

In this paper we consider the problem of sequencing of both arrivals and departures on a single runway. In Section 2 we describe the problem with its characteristics and constraints. In Section 3 we present a Mixed Integer Linear Programming (MILP) formulation and a rolling horizon solution approach. In Section 4, computational results on real world air traffic instances from the Milano Linate Airport are reported. Finally, in Section 5, we report conclusions and further developments.

## 2 Problem description

The air traffic controllers are responsible to provide a minimum separation between arrivals, between departures and between arrivals and departures. The minimum separation depends on several factors; the most important are the wake turbulence, the in and outbound routes, the aircraft types and performances. Arriving aircraft leave the en-route airspace via the Standard Arrivals Route (STAR) and reach the Initial Approach Fix Point where they start the final approach procedure. The Expected Approach Time (EAT) is the time at which an inbound aircraft will be cleared for the final procedure.

Once started the final approach, changes in the landing sequence can be obtained only with a big workload for the controllers. Delay actions are realistic only before this phase. Moreover, the aircraft speeds during the final approach phase are similar and can be approximated with a constant value. In this way it is possible to define an Estimated Landing Time (ELT) for each arrival.

Once terminated the boarding operations, the pilots of a departing aircraft request the startup clearance to the Tower Controller. If the delay foreseen at the holding point is acceptable, the startup is approved and the aircraft is instructed to taxi (or possibly to a previous push back). The taxi time from the parking to the holding point (holding position before entering the runway) can be approximated with

the value of the airport average taxi time. In this way it is possible to define an Estimated Take-Off Time (ETT). Delay actions for a departure are possible before the engines start up (at the parking). After that moment delay actions are not efficient from the fuel consumption perspective. Even though the Calculated Take-Off Time system (defined from the Central Flow Management Unit of Eurocontrol)

is able to respect the capacity of many airports, the problem of sequencing the arrivals and the departures remain a tactical problem for the Air Traffic Controllers. Currently the delay actions used to tactically define the sequence consist of a "long vectoring", holding procedures or speed control for an arrival, and in a holding procedure on the taxiways near the holding point for a departure. One of the main ideas of the future Air traffic Management (ATM) procedures consists of assigning a delay at the parking (before the engine start) or before the beginning of the final procedure (delay on the EAT, the aircraft properly reduces the speed during the en-route phase). In this context a sequencing algorithm is used to identify the Target Take-off Time (TTOT) and the Target Landing Time (TLT) that represent the runway slots for the users (departures and arrivals). Once the taxi time is known, the approach time and other parameters, as the runway pressure, a Target Start up Approval Time (TSAT) and a Target Expected Approach Time (TEAT), are provided to both outbound and inbound traffic. It is important to note that this sequencing problem is dynamic and belongs to a specific time window. In other words, the sequencing horizon does not correspond to the actual time, i.e. the sequence in the present moment has been already defined and cannot be changed with an acceptable workload for the controllers. The sequence that can be decided is the sequence of arrivals and departures that will interest the runway after a given time period (for example 10 minutes). On the other hand, many dynamic approaches to the aircraft sequencing problem perform a sequence re-planning once a new aircraft enters the approach airspace. The first consequence is that, in order to comply with real time applications, a fast heuristic

approach is often used. One of the most critical problems of this methodology is the sequence consistency, i.e. after each re-planning it is important to check that the changes in the sequence are realistic and acceptable from the controllers' perspective. For those reasons we propose a rolling horizon approach to the problem. If  $t$  is the actual time, we consider two time windows: in the first one, of duration  $\Delta T_f$ , the sequence is frozen, while the second window of length  $\Delta T$ , represents the optimization horizon. The sequencing problem is solved considering only the aircraft in the optimization horizon; then at each iteration the optimization horizon "rolls" forward. In details in this work the windows of the optimization horizon is modeled using a fixed number of aircraft.

## 2.1 Aircraft separation and operative constraints

In the controlled airspace near big airport (classified as airspace A by the International Civil Aviation Organization ICAO) a separation must be provided between each couple of aircraft. Different kinds of separations can be applied; the most important are the RADAR separation (longitudinal separation of 5 NM, 3NM in congested area, and vertical separation of 1000ft) and the wake turbulence separation. The latter separation depends on the aircraft dimensions and weight. For this scope the ICAO defines four categories of wake turbulence: Light (L), Medium(M), Heavy(H) and Super (only the Airbus A380).

The wake turbulence separation must be applied between arrivals, between departures and between arrivals and departures if it is foreseen an intersection between the aircraft trajectories (for example in case of departure from an intermediate position). Moreover, if the controllers have the aircraft in sight, they can properly reduce the separation between them. This is what happens between an arrival and a departure during normal weather conditions. For these reasons it is hard to model separations between arrivals and departures in a realistic way. To overcome this problem we consider the case of the normal weather conditions without departures from intermediate positions; we suppose that one minute is sufficient to the inbound aircraft to vacate the runway and to the departing aircraft to free the runway.

Another important consideration is related to the different priorities in the landing and take-off

**Table 1:** Notation:

|                               |                                                                 |
|-------------------------------|-----------------------------------------------------------------|
| $i \in I$                     | : set of aircraft                                               |
| $k \in K = \{1, \dots,  I \}$ | : set of possible positions of sequence $s$                     |
| $t_I^i$                       | : scheduled arrival\departure time associated with aircraft $i$ |
| $t_S^{i_1 i_2}$               | : minimum separation between aircraft $i_1$ and $i_2$           |
| $w_i$                         | : cost of a time unit delay for aircraft $i$                    |
| $d_i$                         | : maximum delay acceptable for aircraft $i$                     |
| $p$                           | : maximum position shift                                        |

operations. Operative practice assigns the priority to the inbound traffic. It is not realistic that a holding minute on the ground has the same weight of a minute of flight, i.e. an arriving aircraft cannot be delayed as a departing one. Finally an important constraint to be considered is the so called constrained shift positioning: i.e. the position of an aircraft in sequence can be realistically shifted by a reduced number of positions respect to the First Come First Served (FCFS) sequence.

## 2.2 Relationships with other optimization problems

It is important to note that the problem of sequencing aircraft on the runway can be modeled as a Machine Scheduling Problem and as a Traveling Salesman Problem with Time Windows (TSPTW). Brentnall [2] underlined the relationship between the ALP and a machine scheduling problem with sequence dependent setup times. Each job corresponds to a landing operation of an aircraft, each machine with capacity one represents a runway, the estimated landing time (ELT) of an aircraft corresponds to the release date of the corresponding job and the actual landing time (ALT) corresponds to the starting time. Moreover, the time when an aircraft vacates the runway corresponds to the completion time of the related job and the separation between the aircraft represents the sequence-dependent processing time between jobs. In the same way the relationship with the TSPTW (reported by Luenberger [16]) is relevant: each aircraft corresponds to a city, the distance between the cities corresponds to the separation between the aircraft and the landing time window defines the time window of the traveling salesman.

## 3 A MILP formulation of the ASP

Let us formalize the problem and introduce some notation (resumed in Table 1). Let  $I$  be the set of aircraft (arrivals and departures) that will use the runway during the rolling horizon  $\Delta T$ . For each aircraft  $i \in I$ , let  $t_I^i$  be the scheduled time to use the runway, and  $d_i$  the maximum acceptable delay. The weight of a time unit delay for aircraft  $i$  is equal to  $w_i$ . For each ordered pair of aircraft  $(i_1, i_2)$ , ( $i_1 \in I, i_2 \in I$ ) a minimum separation  $t_S^{i_1 i_2}$  must be applied. Each aircraft of the sequence can be shifted by at most  $p$  positions with respect to the FCFS order. The goal is to find the sequence of aircraft that minimizes the total weighted delay during the rolling horizon  $\Delta T$ . We assume that all the input data are positive numbers.

The main idea of the following model is to construct a feasible sequence consisting of  $|I|$  positions to which aircraft can be assigned. Let  $k \in K = \{1, \dots, |I|\}$  be the set of possible positions in a sequence of aircraft. We define the following sets of decision variables: a binary variable  $x_k^i$  which takes value 1 if  $i$  ( $\in I$ ) is assigned to position  $k$  ( $\in K$ ), a continuous variable  $y_k$  which represents the arrival\departure time of the aircraft in position  $k$  ( $\in K$ ) of the sequence and finally a continuous variable  $z^i$  which represents the arrival\departure time for aircraft  $i$  ( $\in I$ ).

The mathematical formulation of the problem is given in Model ((1)-(8)). The objective function (1) aims at minimizing the total cost caused by the delay of a sequence.

$$\text{Min} \quad \sum_{i \in I} w_i(z^i - t_I^i) \quad (1)$$

$$\text{s. t.} \quad \sum_{k \in K} x_k^i = 1 \quad i \in I \quad (2)$$

$$\sum_{i \in I} x_k^i = 1 \quad k \in K \quad (3)$$

$$z^i \leq t_I^i + d_i \quad i \in I \quad (4)$$

$$z^i \geq y_k + M(x_k^i - 1) \quad k \in K, i \in I \quad (5)$$

$$y_k \geq \sum_{i \in I} t_I^i x_k^i \quad k \in K \quad (6)$$

$$y_k \geq y_{k-1} + t_S^{i_1 i_2} (x_{k-1}^{i_1} + x_k^{i_2} - 1) \quad k \in \bar{K}, (i_1, i_2) \in I \quad (7)$$

$$x_k^i \in \{0, 1\}, y_k \in \mathbb{R}^+, z^i \in \mathbb{R}^+ \quad i \in I, k \in K. \quad (8)$$

Constraints (2) and (3), ensure respectively that each aircraft must be assigned to one position in a sequence and in each position there must be one aircraft. The delay of aircraft  $i$  must respect its upper bound (constraints (4)). The "big-M" constraints (5) impose that the arrival\departure of an aircraft is equal to the the arrival\departure of a position if and only if the aircraft is assigned to that specific position. No arrival\departure from a position  $k$  is possible before the scheduled arrival\departure time of the assigned aircraft (constraints (6)). We define  $\bar{K} = K \setminus \{1\}$ , constraints (7) impose that the departure\arrival time of specific position in the sequence is not smaller than the departure\arrival time of the previous position plus the separation time of the assigned aircraft. Moreover constraints (6)-(7) together ensure that  $z_i \geq t_I^i$ . Constraints (8) define the allowable values for the decision variables. Each aircraft  $i$  is allowed to shift from its original position in the sequence, say  $b_i$  (defined by the FCFS order), of at most  $p$  positions. This is imposed by setting to zero the corresponding variables  $x_k^i$ , ( $k = b_i + p + 1, \dots, |I|$ ) and ( $k = 1, \dots, b_i - p - 1$ ).

The following inequalities impose an initial condition on the delay by considering the last aircraft  $i_I$  assigned before the optimized sequence and the best possible initial time  $t_{in}$ :

$$y_1 \geq t_S^{i_I i} x_1^i + t_{in} \quad i \in I. \quad (9)$$

Moreover in order to break some symmetry we can impose the following order on the variables  $y$ :

$$y_k \geq y_{k-1} \quad k \in \bar{K}. \quad (10)$$

Constraints (10) help in strengthening the Linear Programming (LP) relaxation of the model. To deal with long aircraft sequences, we propose a Rolling Horizon Algorithm (RHA) which solves iteratively problem (1)-(8) for a variable time window, defined by a predetermined number of aircraft. This time window rolls forwards at each iteration. The last aircraft of the current iteration represents the initial condition for the following one. The RHA starts considering no delay as initial condition for the first iteration and it stops when the current rolling horizon time window includes the last aircraft of the entire sequence to be optimized.

**Table 2:** Instance features.

| Instance | FCFS | departures | arrivals | Light | Medium | begin time | end time |
|----------|------|------------|----------|-------|--------|------------|----------|
| FPT01    | 401  | 30         | 30       | 3     | 57     | 14 55      | 17 14    |
| FPT02    | 457  | 30         | 30       | 4     | 56     | 15 30      | 18 15    |
| FPT03    | 393  | 32         | 28       | 4     | 56     | 15 47      | 18 15    |
| FPT04    | 480  | 29         | 31       | 3     | 57     | 16 14      | 18 16    |
| FPT05    | 440  | 25         | 35       | 4     | 56     | 16 35      | 19 16    |
| FPT06    | 316  | 30         | 30       | 8     | 52     | 14 00      | 16 14    |
| FPT07    | 341  | 29         | 31       | 7     | 53     | 14 32      | 17 14    |
| FPT08    | 306  | 26         | 34       | 9     | 51     | 14 55      | 17 14    |
| FPT09    | 296  | 28         | 32       | 6     | 54     | 15 25      | 18 15    |
| FPT10    | 330  | 28         | 32       | 5     | 55     | 15 55      | 18 15    |
| FPT11    | 354  | 23         | 37       | 5     | 55     | 16 24      | 19 16    |
| FPT12    | 269  | 23         | 37       | 5     | 55     | 16 45      | 20 16    |

**Table 3:** Aircraft weights

|                           |   |                            |   |
|---------------------------|---|----------------------------|---|
| Airbus A320               | 5 | Fokker F70                 | 4 |
| Airbus A319               | 4 | Fokker F100                | 5 |
| Airbus A321               | 6 | Canadian Regional Jet CRJ7 | 4 |
| Mcdonnell Douglas MD80/82 | 6 | Canadian Regional Jet CRJ9 | 5 |
| Boeing B737-600           | 5 | Cessna C550                | 1 |
| Boeing B737-800           | 6 | LearJet L30/50/60          | 2 |
| Boeing B737-500           | 4 | Hawker H25B                | 2 |
| British Aerospace BA46    | 4 | Falcon F900                | 2 |

## 4 Computational experiments

### 4.1 The real air traffic instances

We test our approach using 12 instances obtained from the real air traffic of the Milano Linate Airport (ICAO code LIML). This is the biggest Italian airport for the commercial traffic with a single runway; the second runway is used for the general air traffic. The aerodrome chart is presented in Figure 1, the main parking for commercial aviation is the north apron. From their stands the departing aircraft taxi to the Holding Point Runway 36 via T taxiway. The average taxi time varies during the day accordingly to the peak hours: 10 minutes during non congested hours up to 25 minutes in low visibility conditions during peak hours. We assume an average taxi time of 10 minutes. The arriving aircraft vacate the runway via taxiway G and proceed directly to their parking stand. The time necessary to reach the parking is not influent in the sequencing problem. Typically the runway handles an average of 450 movements per day. We consider two simulation days: one in August 2011 (from 14:00 to 20:00 UTC) and one in September 2011 (from 15:00 to 21:00 UTC). For each day, six instances have been created (FPT01-12) considering 60 aircraft per instance. The weights of the aircraft are obtained considering the number of seats and a medium value of the fuel consumption. In particular we multiply the hundreds of seats times the Ton/hour of fuel and rounded up the results. Table 3 reports the value of the weights used for the most common aircraft. The ICAO separation used between each couple of aircraft and the instances are available on line: [www.or.deis.unibo.it/research.html](http://www.or.deis.unibo.it/research.html). In particular, for each aircraft we report the wake turbulence category, the type of operation (Arrival or Departure), the Estimated Landing Time or the Estimated Take-off Time (as sum of the Estimated of Block Time and of the average taxi time).

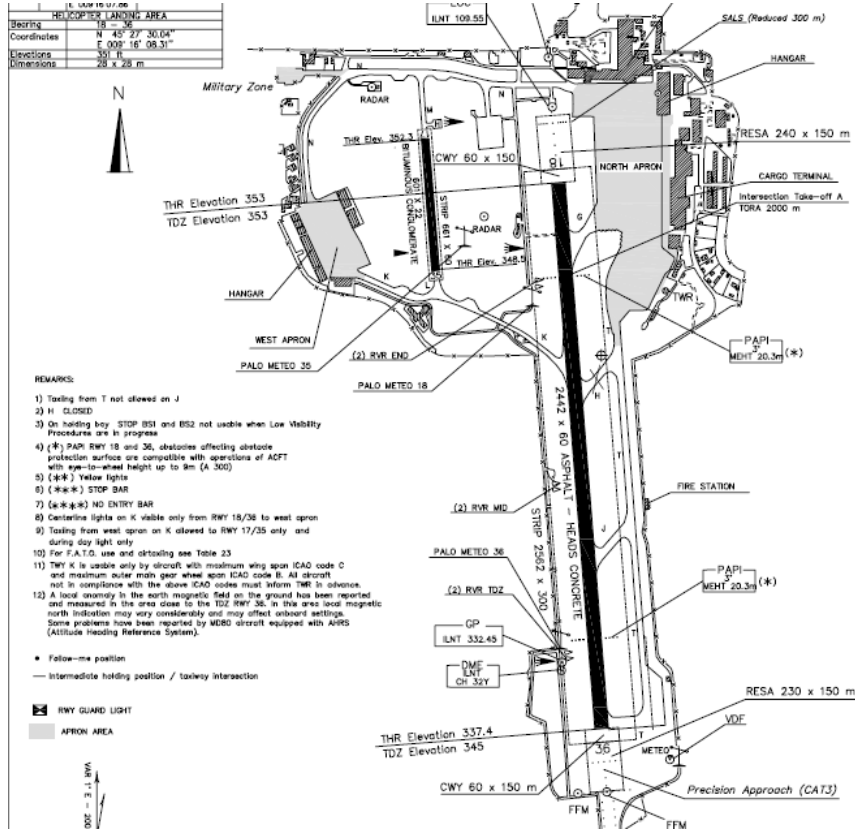


Figure 1: Milano Linate Airport aerodrome chart

## 4.2 Results of the rolling horizon algorithm

Each optimization of the RHA is performed on overlapping sequences of a given dimension and the output of an optimized sequence becomes the initial condition of the following one until all the aircraft of the instance are optimized. For each instance we performed 9 tests grouped in three configurations. For each configuration we consider 3 different dimensions of the rolling horizon aircraft sequence (5 - 10 - 15 aircraft). The first configuration does not impose any shift position constraints. The second and the third configurations have the same sequence dimension but impose, respectively, a maximum shift position constraint of 4 and 8 positions. In our test instances we do not impose any upper bound on the aircraft delay and we consider the same weight for inbound and out bound traffic. All the results of our algorithm were obtained on one core of a Pentium 4 at 3.00GHz with 1Gb RAM under Linux operating system.

In the experiments, we used CPLEX12.3 as linear programming solver. Constraints (5) are imposed using the indicator constraints of CPLEX, taking the place of the "big-M" ones. In Table 2 (Instance features) we report the instance name, the cost computed with the first come first served algorithm (FCFS) and the percentage cost improvement obtained by the proposed algorithm with respect to the FCFS sequence. Then we report some features of the instances. In detail, we show the number of departing and arriving aircraft, the number of light and medium aircraft (no heavy aircraft was present in the Milano Linate Airport during the considered days) and finally the complete time horizon. In Table 4 (Algorithm performance details) we report for each instance and for each configuration the minimum cost obtained by the three runs of the

**Table 4:** Algorithm performance details.

| Instance | Configuration 1 |      |       | Configuration 2 |      |       | Configuration 3 |      |       |
|----------|-----------------|------|-------|-----------------|------|-------|-----------------|------|-------|
|          | best            | time | impr  | best            | time | impr  | best            | time | impr  |
| FPT01    | 329             | 185  | 17.96 | 359             | 7    | 10.47 | 300             | 193  | 25.19 |
| FPT02    | 393             | 199  | 14.00 | 402             | 7    | 12.04 | 344             | 193  | 24.73 |
| FPT03    | 349             | 183  | 11.20 | 354             | 8    | 9.92  | 318             | 187  | 19.08 |
| FPT04    | 395             | 205  | 17.71 | 431             | 7    | 10.21 | 357             | 205  | 25.63 |
| FPT05    | 347             | 220  | 21.14 | 397             | 7    | 9.77  | 331             | 207  | 24.77 |
| FPT06    | 272             | 180  | 13.92 | 275             | 7    | 12.97 | 239             | 189  | 24.37 |
| FPT07    | 250             | 197  | 26.69 | 273             | 7    | 19.94 | 249             | 184  | 26.98 |
| FPT08    | 208             | 189  | 32.03 | 233             | 7    | 23.86 | 193             | 196  | 36.93 |
| FPT09    | 211             | 184  | 28.72 | 235             | 7    | 20.61 | 218             | 190  | 26.35 |
| FPT10    | 237             | 193  | 28.18 | 266             | 8    | 19.39 | 232             | 187  | 29.70 |
| FPT11    | 259             | 193  | 26.84 | 290             | 6    | 18.08 | 267             | 192  | 24.58 |
| FPT12    | 231             | 207  | 14.13 | 232             | 6    | 13.75 | 212             | 241  | 21.19 |
| min      |                 | 180  | 11.20 |                 | 6    | 9.77  |                 | 184  | 19.08 |
| max      |                 | 220  | 32.03 |                 | 8    | 23.86 |                 | 241  | 36.93 |
| avg      |                 | 195  | 21.04 |                 | 7    | 15.09 |                 | 197  | 25.79 |

algorithm, the total computing time (seconds) and the percentage cost improvement obtained by the proposed configuration with respect to the FCFS sequence. Configuration 1 have an average improvement of 21.04% (minimum improvement 11.20% and maximum improvement 32.03%), configuration 2 have an average improvement of 15.09% (minimum improvement 9.77% and maximum improvement 23.86%) and finally configuration 3 have an average improvement of 25.79% (minimum improvement 19.08% and maximum improvement 36.93%).

Finally in table 5 we report an example of solution for instance FPT01 in configuration 3 using a rolling horizon sequence of 10 aircraft. In the first column we report the FCFS sequence, the aircraft type, the turbulence category, the type of operations (Arrival or Departure) and the ideal scheduled time. In the last two columns we report the RHA optimized aircraft sequence with the real scheduled time.

Our experiments showed that the computational time grows exponentially with the length of the sequence to be optimized and it becomes too high for practical purposes for a rolling horizon sequence of more than 20 aircraft. Furthermore, practitioners in this field suggest to impose a time limit for the algorithm of at most 5 minutes to plan a time horizon of 60 aircraft. Accordingly for configurations 1 and 3 we set a maximum time limit of 300 seconds, divided for the number of sequences. If a specific MILP of a sequence reaches the time limit we interrupt the optimization using the best solution computed. Configuration 2 is designed to be faster the others, specifically thought for the cases in which a real time re-optimization is needed, but at the same time guaranteeing good performance results.

## 5 Conclusions and further developments

The runway is one of the most critical bottlenecks of the air transportation system; aircraft scheduling on the runway is a challenging problem that aims to reduce the delays and the controllers' workload. In this paper we consider the problem of sequencing both arrivals and departures; we formalize the problem and propose a MILP model. A rolling horizon approach has been used and tested on real air traffic data from the Milano Linate Airport. The results

show that this approach is able to significantly improve on the FCFS sequence. Further developments will focus on the comparison with other algorithms proposed in the ASP literature, and in the research of valid inequalities from the machine scheduling literature. The final objective consists of the design of a supporting tool for air traffic controllers that is able to suggest a valid runway utilization sequence.

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**Table 5:** Optimization results of FPT01 instance.

| FCFS    | Type | Cat | A/D | time  | RHA     | time  |
|---------|------|-----|-----|-------|---------|-------|
| VLE7100 | A320 | M   | D   | 14 55 | VLE7100 | 14 55 |
| VLE7096 | A320 | M   | D   | 14 55 | VLE7096 | 14 57 |
| AZA2061 | A320 | M   | D   | 15 00 | AZA2061 | 15 00 |
| ONG254  | H25C | M   | A   | 15 02 | ONG254  | 15 02 |
| ISS1508 | MD82 | M   | D   | 15 05 | ISS1508 | 15 05 |
| SMX5297 | A319 | M   | D   | 15 15 | SMX5297 | 15 15 |
| VLE7110 | MD82 | M   | A   | 15 15 | VLE7110 | 15 17 |
| CYL5001 | A320 | M   | A   | 15 18 | CYL5001 | 15 19 |
| AZA410  | E170 | M   | D   | 15 25 | VLE7088 | 15 25 |
| VLE7088 | A319 | M   | D   | 15 25 | AZA410  | 15 27 |
| AZA2063 | A320 | M   | D   | 15 30 | AZA2063 | 15 30 |
| NJE210A | C56X | M   | A   | 15 31 | NJE210A | 15 32 |
| CYL5022 | A320 | M   | A   | 15 34 | CYL5022 | 15 34 |
| AZA080  | A320 | M   | D   | 15 35 | AZA080  | 15 36 |
| BAW14GM | A319 | M   | A   | 15 35 | BEL7PC  | 15 38 |
| ICFLY   | LJ31 | M   | A   | 15 35 | BAW14GM | 15 40 |
| ISS1205 | A319 | M   | A   | 15 37 | ISS1205 | 15 42 |
| SMX5985 | A320 | M   | A   | 15 37 | ICFLY   | 15 44 |
| BEL7PC  | RJ85 | M   | A   | 15 42 | SMX5985 | 15 46 |
| SNM643  | F2TH | M   | A   | 15 47 | AZA2092 | 15 48 |
| AZA2092 | A319 | M   | A   | 15 47 | DLH1XK  | 15 50 |
| DLH1XK  | B735 | M   | D   | 15 50 | SNM643  | 15 52 |
| ISS223  | MD82 | M   | D   | 15 55 | ISS223  | 15 55 |
| AZA79A  | A320 | M   | D   | 15 55 | AZA79A  | 15 57 |
| DEPPG   | PA46 | L   | A   | 15 58 | ICAFD   | 16 00 |
| SNM834  | F900 | M   | D   | 16 00 | AZA2097 | 16 02 |
| ICAFD   | F50  | M   | D   | 16 00 | SNM834  | 16 04 |
| AZA2097 | A320 | M   | D   | 16 00 | DEPPG   | 16 06 |
| AZA358  | A319 | M   | D   | 16 10 | AZA358  | 16 10 |
| VLE7046 | A320 | M   | A   | 16 13 | VLE7046 | 16 13 |
| KLM1621 | B738 | M   | A   | 16 14 | KLM1621 | 16 15 |
| AZA1647 | MD82 | M   | D   | 16 15 | AZA1647 | 16 17 |
| AZA2088 | A320 | M   | A   | 16 17 | AZA2088 | 16 19 |
| AZA2099 | A320 | M   | D   | 16 20 | AZA2099 | 16 21 |
| SNM833  | F900 | M   | A   | 16 26 | SNM833  | 16 26 |
| ADH2913 | A320 | M   | A   | 16 28 | ADH2913 | 16 28 |
| SNM644  | F2TH | M   | D   | 16 30 | DLH276  | 16 30 |
| OOELI   | D328 | M   | D   | 16 30 | SNM644  | 16 32 |
| DLH276  | A321 | M   | A   | 16 31 | ISS1206 | 16 35 |
| ISS1206 | A320 | M   | D   | 16 35 | OOELI   | 16 37 |
| AZA1727 | A319 | M   | D   | 16 35 | VLE7008 | 16 39 |
| VLE7008 | MD82 | M   | A   | 16 39 | VLE7048 | 16 41 |
| AZA2DJ  | A320 | M   | D   | 16 40 | AZA2DJ  | 16 43 |
| VLE7048 | CRJ9 | M   | A   | 16 41 | AZA1727 | 16 45 |
| ADH6856 | A320 | M   | D   | 16 45 | ADH6856 | 16 47 |
| AZA2050 | A319 | M   | A   | 16 49 | AZA2050 | 16 49 |
| BEL10D  | RJ85 | M   | D   | 16 50 | BEL10D  | 16 51 |
| ICDBS   | C56X | M   | A   | 16 52 | ICDBS   | 16 53 |
| AOE402  | F2TH | M   | D   | 17 00 | VLE7101 | 17 00 |
| AZA2109 | A319 | M   | D   | 17 00 | AOE402  | 17 02 |
| AUF413  | C56X | M   | D   | 17 00 | AZA9HM  | 17 04 |
| ADH7VG  | A320 | M   | A   | 17 01 | ADH7VG  | 17 06 |
| VLE7101 | A321 | M   | A   | 17 02 | AUF413  | 17 08 |
| AZA9HM  | A320 | M   | A   | 17 08 | AZA2109 | 17 10 |
| BAW47NL | A319 | M   | D   | 17 10 | BAW47NL | 17 12 |
| SMX5610 | A320 | M   | D   | 17 15 | SMX5610 | 17 15 |
| AFR2112 | A320 | M   | A   | 17 22 | AFR2112 | 17 22 |
| AZA2102 | A319 | M   | A   | 17 27 | AZA2102 | 17 27 |
| INGIR   | PRM1 | L   | A   | 17 28 | YUBZZ   | 17 30 |
| YUBZZ   | C550 | L   | A   | 17 30 | INGIR   | 17 32 |

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